

ABHISHEK DWIVEDI

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PROFESSIONAL SUMMARY

Bioinformatics postgraduate (M.Sc, Central University of Himachal Pradesh, CGPA 8.33/10.0) with hands-on experience in computational drug discovery, molecular dynamics, genomics data analysis, and machine learning. Built and published six end-to-end computational pipelines covering molecular docking and QM/MM calculations, steered and replica-exchange molecular dynamics, NGS variant calling, RNA-seq quality control, and semantic gene-pathway analysis using BioBERT. Seeking a role in bioinformatics analysis, computational drug discovery, or genomics R&D.

CORE COMPETENCIES

Molecular Docking & CADD • Molecular Dynamics (GROMACS, OpenMM) • QM/MM & DFT (ORCA) • NGS Variant Calling (GATK, BWA) • RNA-seq & 16S rRNA Analysis • Machine Learning (Python, Scikit-learn) • BioBERT / NLP for Genomics • Python, R, SQL, Bash • Docker & Reproducible Pipelines

EDUCATION

Central University of Himachal Pradesh

M.Sc. in Bioinformatics | CGPA: 8.33/10.0

Aug 2024 – 2026

Amity University, Lucknow Campus

B.Sc. in Biotechnology | CGPA: 9.02/10.0

Jul 2019 – Jun 2022

M.SC. DISSERTATION

Comparative Genomic Analysis of Antimicrobial Resistance in *Salmonella enterica*

2026

Supervisor: Dr. Dinesh Lakhanpal, Centre for Computational Biology & Bioinformatics, CUHP

- Conducted a comparative genomic study of 67 complete *Salmonella enterica* genomes from European clinical isolates, designing a fully Dockerized, reproducible workflow spanning MLST, phylogenetics, AMR gene profiling, virulence profiling, and mobile genetic element (MGE) detection.
- Containerized each pipeline stage as an isolated Docker image (mlst v2.32.2, MAFFT/FastTree, RGI v6.0.3 with CARD, ABRicate v1.0.1 with VFDB, MobileElementFinder v1.0.2) to guarantee consistent software versions and reproducibility across the full genome batch.
- Identified 22 distinct MLST sequence types (dominant lineages: ST11 *S. Enteritidis*, ST198 *S. Kentucky*), built a 16S rRNA maximum-likelihood phylogeny (1,000 bootstrap replicates), and profiled the resistome, virulome, and mobilome, detecting the colistin-resistance gene *mcr-9.1* and quantifying IS/MITE-driven resistance dissemination (100% and 43.3% prevalence, respectively).
- Presented findings as a poster at the International Conference on Environmental Sustainability, Climate Adaptation & Disaster Management (ESCAD-2026), Central University of Himachal Pradesh.

PROJECTS

NGS Variant-Calling Pipeline for *Mycobacterium tuberculosis* [\[GitHub ↗\]](#)

- Built a three-stage, reference-based NGS pipeline in Bash (GATK 4.5, BWA, samtools, bcftools, SnpEff, maftools) covering per-sample QC/alignment/variant calling, cohort-level joint genotyping, and MTB-specific annotation.
- Processed two public MTB whole-genome samples end-to-end, achieving 99.5–99.8% genome coverage and identifying 1,875 non-silent variants across 901 mutated genes, consistent with known PE/PPE hypervariable loci.

QM/MM Toolkit for Protein-Ligand Systems [\[GitHub ↗\]](#)

- Developed a config-driven Python toolkit combining OpenMM (molecular mechanics) and ORCA (quantum mechanics) to run QM/MM single-point calculations on protein-ligand complexes.
- Ran a case study on T4 Lysozyme L99A/M102Q bound to ligand JZ4 (B3LYP/def2-SVP, D3BJ), quantifying a 0.34e polarization shift in the ligand's hydrogen-bonding atom caused by the protein environment.

Steered Molecular Dynamics of Protein-Ligand Unbinding [\[GitHub ↗\]](#)

- Designed and ran a GROMACS/CHARMM27 steered MD simulation (92,694 atoms) to pull a ligand out of its binding pocket and quantify the unbinding force profile.
- Identified a peak rupture force of ~475 pN and produced force-time, force-distance, and structural visualizations using Python and PyMOL.

Replica Exchange Molecular Dynamics (REMD) Simulation [\[GitHub ↗\]](#)

- Implemented an 8-replica temperature REMD simulation of a T4 Lysozyme fragment using GROMACS, including full system preparation, equilibration, and production run.
- Wrote a Python analysis pipeline to evaluate replica exchange diagnostics (potential-energy overlap, transition matrices, exchange probabilities) and diagnose ladder-spacing issues.

Gene-to-Pathway Semantic Similarity Pipeline [\[GitHub ↗\]](#)

- Built an NLP-based pipeline using BioBERT sentence embeddings and cosine similarity to match upregulated genes to Reactome pathways beyond formal gene-pathway annotations.
- Automated gene annotation (mygene.info API) and pathway summary retrieval (Reactome REST API), visualized as heatmaps and interactive bipartite network graphs.

RNA-seq Quality Control and Preprocessing Pipeline [\[GitHub ↗\]](#)

- Performed end-to-end QC and adapter/quality trimming on public SRA RNA-seq data using FastQC and fastp, validating improvements in per-base quality and duplication metrics before downstream alignment.

CERTIFICATIONS & TRAINING

- Computer-Aided Drug Design (CADD) Master Certification — BioTecNika (Jun 2025)
- Computational Drug Discovery, Docking, MD & QM/MM Approaches — DE<code> LIFE (May 2026)
- National Workshop on WGS & RNA-Seq — BioTecNika (Dec 2025)
- Bioinformatics Methods & Tools Master — BioTecNika (Nov 2024)
- Machine Learning Virtual Internship — CodeAlpha (Aug 2025)
- One-Month International Workshop on Proteins: Sequence, Structure & Function — DE<code> LIFE (Jul 2025)
- AI in Genomics & Healthcare with Python — International Workshop — DE<code> LIFE (Aug 2025)
- Full Stack Development with AI — Physics Wallah (PW) (Jul 2026)

ADDITIONAL EXPERIENCE

Summer Intern, Department of Pathology — Dr. Ram Manohar Lohia Institute of Medical Sciences*Mar – Apr 2022*

- Completed hands-on training analyzing Levey-Jennings charts for coagulation test quality control, gaining exposure to hematology lab workflows and diagnostic laboratory operations.

Co-Author, Review Paper — Amity University, Lucknow Campus

May 2021 – Jul 2022

- Co-authored a review paper on the integration of telemedicine in COVID-19 management, examining its role in improving healthcare accessibility and reducing burden on traditional care systems.

LEADERSHIP & VOLUNTEERING

- Organizing Committee Member, TARMASR-2025 (National Workshop on Research Methodology & Advanced Statistics) — Central University of Himachal Pradesh; coordinated logistics, registrations, and session management.
- Poster Presenter, National Conference on Pollution Control & Sustainable Environment — Amity University Lucknow Campus.
- Event Coordinator, DBT-CTEP Popular Lecture Series in Biotechnology — Amity University Lucknow Campus.